

(Please write your Exam Roll No. immediately)

Exam Roll No: | | | | | | | | | | | | | | | |

END TERM EXAMINATION
THIRD SEMESTER [B.TECH.] JANUARY, 2024
Paper Code: CIC-205 • Subject: Discrete Mathematics

Time: 3 Hours

Maximum Marks: 75

Note: Attempt **five** questions in all including **Q. No. 1** which is compulsory. Select **one** question from each unit. Assume missing data, if any.

Q. No.	Question	Max. Marks	CO(s)
Q1 (a)	Define a power set and give an example.	[2]	CO1
Q1 (b)	State De Morgan's Laws in propositional logic.	[2]	CO1
Q1 (c)	What is the principle of mathematical induction?	[2]	CO1
Q1 (d)	Give an example of an equivalence relation.	[2]	CO2
Q1 (e)	Define a Boolean Algebra.	[2]	CO3
Q1 (f)	State the difference between a group and a monoid.	[2]	CO3
Q1 (g)	What is Euler's formula in graph theory?	[2]	CO4
Q1 (h)	Define a Hamiltonian circuit in a graph.	[1]	CO4
UNIT – I			
Q2 (a)	Construct a Venn diagram for three sets A, B, C with the conditions: $A \cap B = \emptyset$, $B \cap C \neq \emptyset$, and $A \cap C \neq \emptyset$.	[5]	CO1
Q2 (b)	Create a truth table for the proposition $\neg(q \wedge r)$ and determine its logical equivalence.	[5]	CO1
Q2 (c)	Define a binary relation R on the set $\{1, 2, 3, 4\}$ where R contains (a, b) if and only if a divides b. Determine if R is reflexive, symmetric, and transitive, providing justification for each property.	[5]	CO2
OR			
Q3 (a)	Using the principle of inclusion and exclusion, find the number of integers between 1 and 100 that are divisible by 2, 3, or 5. Provide step-by-step calculations.	[5]	CO1
Q3 (b)	Prove that in a group of six people, at least three people either all know each other or none of them knows the others. Use the pigeonhole principle in your proof.	[5]	CO1
Q3 (c)	Given a set S of n elements, derive an expression for the number of ways to choose a non-empty subset of S. Validate your expression with $n = 5$.	[5]	CO1
UNIT – II			
Q4 (a)	Let $f(x) = 2x - 1$ and $g(x) = x/2 + 1$. Find the compositions $f(g(x))$ and $g(f(x))$ and determine their domains. Illustrate each step and clarify the reasoning behind the determination of their domains.	[5]	CO2
Q4 (b)	Show that every finite distributive lattice can be represented as a lattice of sets, using the lattice formed by the subsets of the set $\{a, b, c\}$ as an example. Provide a detailed explanation.	[5]	CO2
Q4 (c)	Design a logic circuit for the Boolean expression $(A + B)C + AC$ and simplify it using Karnaugh maps. Include a step-by-step process for the circuit and a detailed explanation of the Karnaugh map simplification.	[5]	CO3
OR			
Q5 (a)	Define Euler's totient function $\phi(n)$. Calculate $\phi(30)$ and explain the significance of each step in your calculation process.	[5]	CO1
Q5 (b)	Demonstrate how generating functions can be used to solve the recurrence relation $a_n = 7a_{n-1} + n$ with $a_0 = 1$, providing a specific example. Illustrate your solution with a clear explanation.	[5]	CO2
Q5 (c)	Apply Master's theorem to analyze the time complexity of the recursive algorithm $T(n) = 2T(n/2) + n \log n$.	[5]	CO2

Q. No.	Question	Max. Marks	CO(s)
UNIT – III			
Q6 (a)	Prove that every element of a finite group has an order. Use this to show that every group of prime order is cyclic. Your proof should be detailed, covering all necessary steps and reasoning.	[5]	CO3
Q6 (b)	Given the group $G = \{1, 2, 3, 4, 5, 6\}$ under multiplication modulo 7, demonstrate how cosets of a subgroup H in G partition G using $H = \{1, 2, 4\}$. Clearly demonstrate and provide a rationale for each step.	[5]	CO3
Q6 (c)	Discuss the significance of normal subgroups in group theory and provide an example of a non-trivial normal subgroup in the symmetric group S_4 . Provide a clear explanation with an illustrative example.	[5]	CO3
OR			
Q7 (a)	Explain the concept of group homomorphism and prove that the kernel of a homomorphism is a subgroup of the domain. Include definitions and a thorough proof with clear logical steps.	[5]	CO3
Q7 (b)	Consider a group $G = \{1, 2, 3, 4, 5, 6\}$ under multiplication modulo 7: (i) Find the multiplication table of G . (ii) Find 2^{-1} , 3^{-1} and 6^{-1} . (iii) Is G cyclic?	[5]	CO3
Q7 (c)	For the symmetric group S_3 , determine all subgroups and identify which are normal. Identify each subgroup and clearly explain why it is or is not a normal subgroup.	[5]	CO3
UNIT – IV			
Q8 (a)	For a given graph G , show how to determine if it has an Eulerian path or circuit. Apply your method to the graph with vertices A, B, C, D and edges AB, BC, CD, DA, AC . Provide a step-by-step methodology for your determination and apply it to the given graph.	[5]	CO4
Q8 (b)	Prove Euler's formula for planar graphs ($V - E + R = 2$) and use this formula to deduce the number of regions in a graph with 5 vertices and 7 edges. Include a detailed proof of Euler's formula and apply it step-by-step to the given graph scenario.	[5]	CO4
Q8 (c)	Using Dijkstra's algorithm, find the shortest path in a weighted graph from A to E . Detail each step in using Dijkstra's algorithm on the provided graph and discuss its time complexity with rationale. [Evaluation Graph: $V = \{A, B, C, D, E\}$, weighted edges: $(A,B)=4, (A,C)=2, (B,C)=1, (B,D)=5, (C,D)=8, (C,E)=10, (D,E)=2$]	[5]	CO4
OR			
Q9 (a)	Describe Kruskal's algorithm for finding a minimal spanning tree (MST) and apply it to a specific graph. Explain each step in Kruskal's algorithm and apply it to the provided graph, detailing the process. Discuss its time complexity.	[5]	CO4
Q9 (b)	Define the chromatic number of a graph. For the cycle graph C_4 composed of vertices A, B, C, D with edges AB, BC, CD, DA , determine the chromatic number. Explain the reasoning behind your determination.	[5]	CO4
Q9 (c)	Discuss the Traveling Salesman Problem (TSP) and illustrate its solution for a set of cities (Delhi, Jaipur, Agra, Chandigarh, Lucknow) beginning from Delhi. Discuss its effectiveness and computational limitations.	[5]	CO4